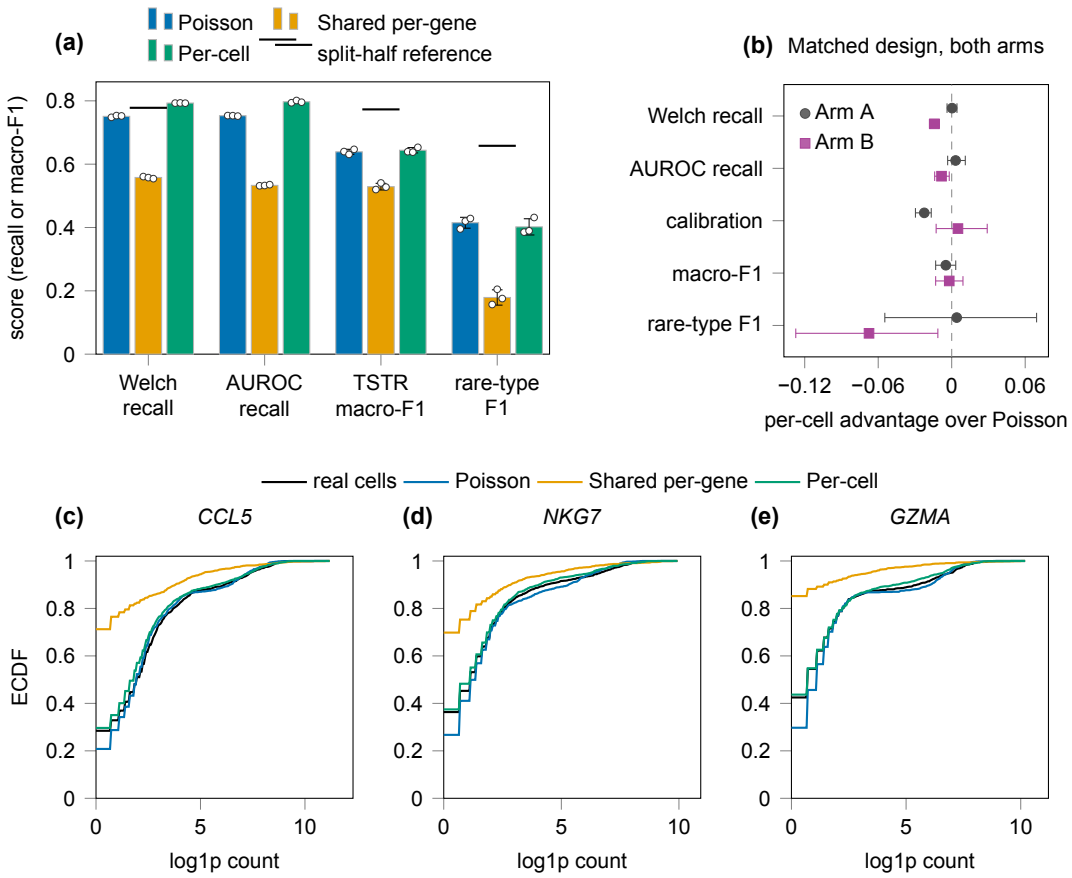

(a)
Poisson
Per-cell
Shared per-gene
split-half reference
score (recall or macro-F1)
0
0.2
0.4
0.6
0.8
Welch recall
AUROC recall
TSTR macro-F1
rare-type F1
(b) Matched design, both arms
Arm A
Arm B
Welch recall
AUROC recall
calibration
macro-F1
rare-type F1
−0.12
−0.06
0
0.06
per-cell advantage over Poisson
real cells
Poisson
Shared per-gene
Per-cell
(c) CCL5
(d) NKG7
(e) GZMA
ECDF
0
0.2
0.4
0.6
0.8
1
0
5
10
log1p count